27/08/2023

$q = 2e^{-4t}$

current ~~whereto~~ when $t = 5\,s$

$i = ?$

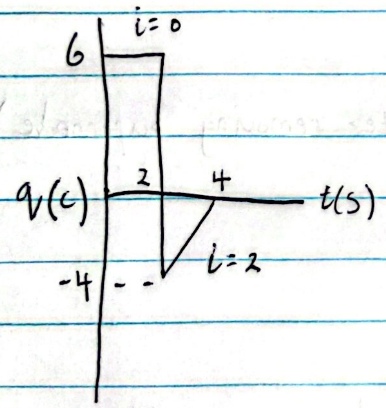
$i = \frac{dq}{dt}$ press shift then integral button to differentiate

$\Rightarrow \frac{d}{dx} 2e^{-4(x)} \Big|_{x=5}$

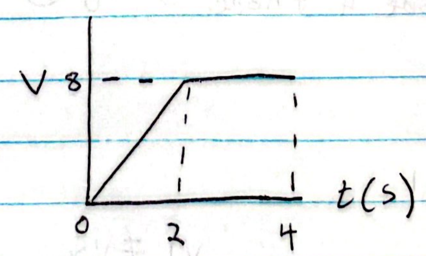
$q = -16{,}49$ nA

use $i = \frac{q}{t}$ when given constant values

$\Rightarrow$ i.e $q = 7\,C$ $t = 6s$



~~calculate power over~~



Calculate power $0 \leq t \leq 4$

vtotal = area

$\frac{1}{2}(2)(8) + (2 \times 8)$

$= 24\ V$

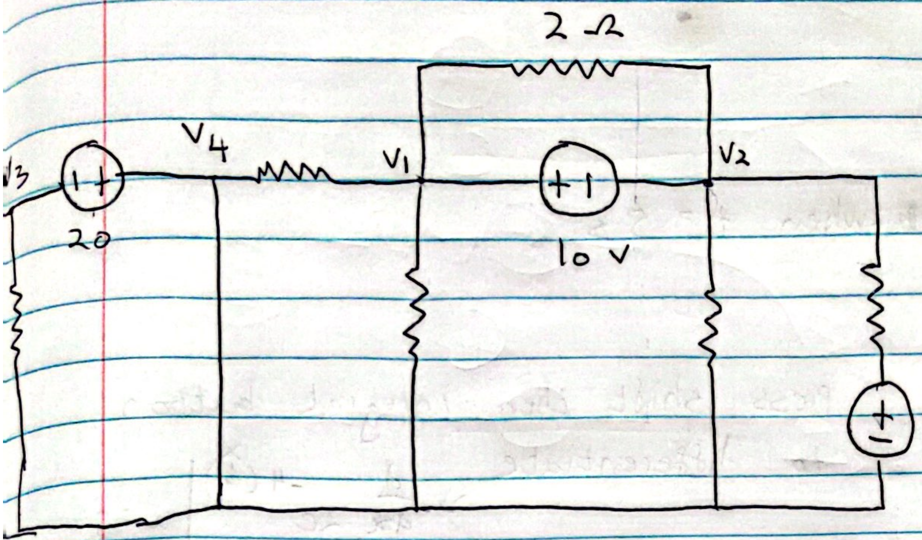
- $i = 0$ $0 \leq t \leq 2$ (gradient)
- $\frac{-4-0}{2-4} = 2$ $i = 2$ $2 < t \leq 4$

i between $0 - 4 = 0 + 2 = 2$

$P = VI$

$= 24 \times 2$

$= 48\ W$



supernode => voltage source between 2 node points without a resistor inbetween

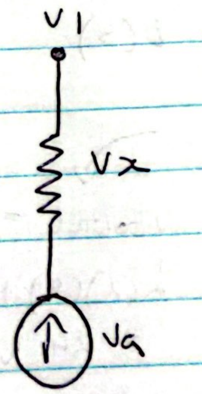
$V1 - V2 = 10$ ①

remove supernode

node 1 + node 2 = 0 ②

$V4 - V3 = 20$ ③

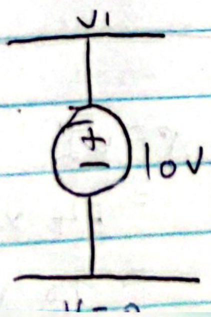
node 4 + node 3 = 0 ④ (after removing supernode)



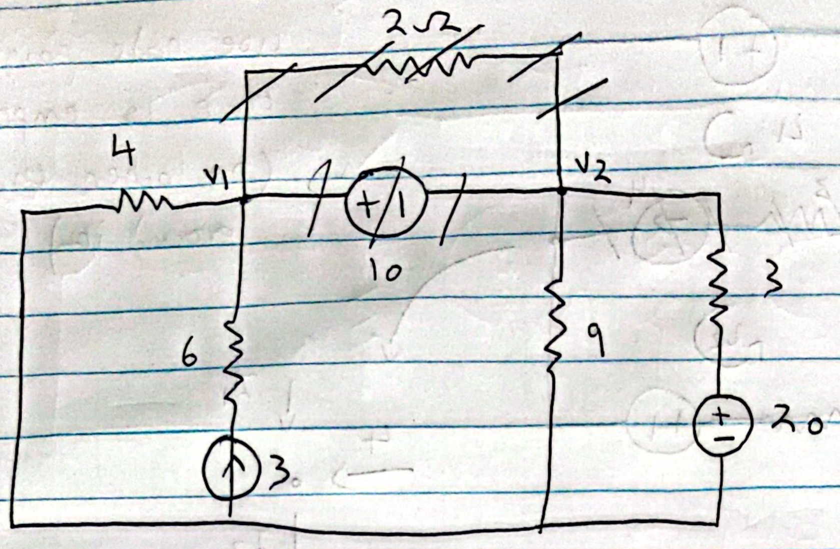
$V1 \neq Vx$

current source has voltage

$V1 = Vx + Va$



$V1 = 10\ V$



• $v_1 - v_2 = 10$

remove supernode AND 2Ω resistor

↓

because it is in parallel w v_1 and v_2

Node 1 + Node 2 = 0

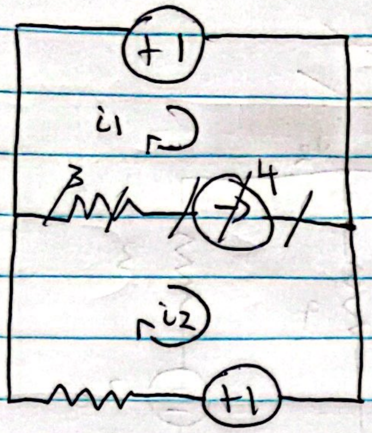
$$\frac{v_1}{4} + (-3) + \frac{v_2}{9} + \frac{v_2 - 20}{3} = 0$$

If you forgot to remove 2Ω resistor

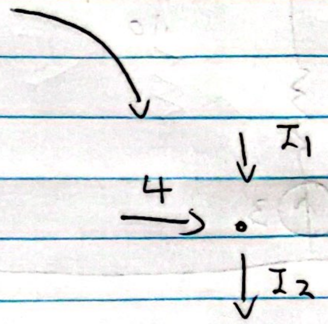
$$\frac{v_1}{4} - 3 + \left[\frac{v_1 - v_2}{2} + \frac{v_2 - v_1}{2}\right] + \frac{v_2}{9} + \frac{v_2 - 20}{3} = 0$$

$0v_1 + 0v_2$

$$\frac{v_1}{4} - 3 + \frac{v_2}{9} + \frac{v_2 - 20}{3} = 0$$



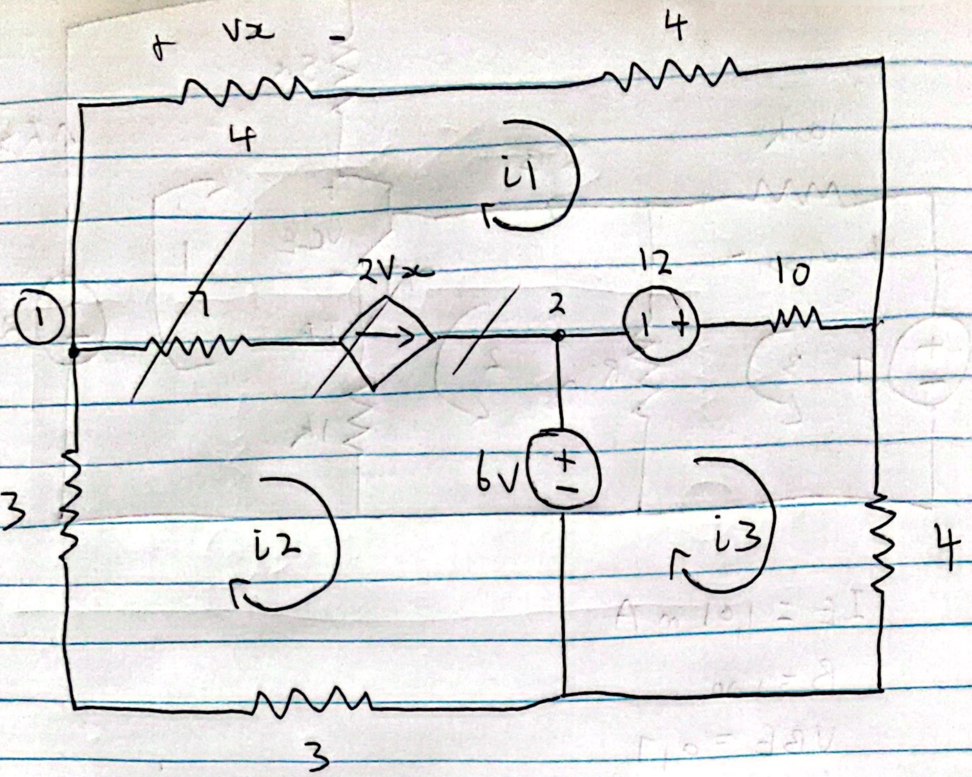
use node point that is empty (no other things around it)

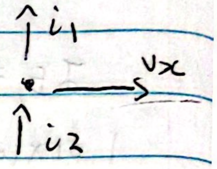


I_n = out

$4 + I_1 = I_2$

when you remove supermesh, also remove the $3\ \Omega$ resistor





② $ai_1 + bi_2 + ci_3 = -18$

① * $di_1 - i_2 = 0$

③ $-10i + 14i_3 = 18$

Point 1

~~loopmh~~ supermesh I_n = out ①

$I_2 = I_1 + 2V_x$

$I_2 = 9I_1$

* $9I_1 - I_2 = 0$ ①

take out supermesh

$V_x = \pm [IR]$

$= + I_1 \times 4$

$V_x = 4I_1$

loop 1 + loop 2 = 0

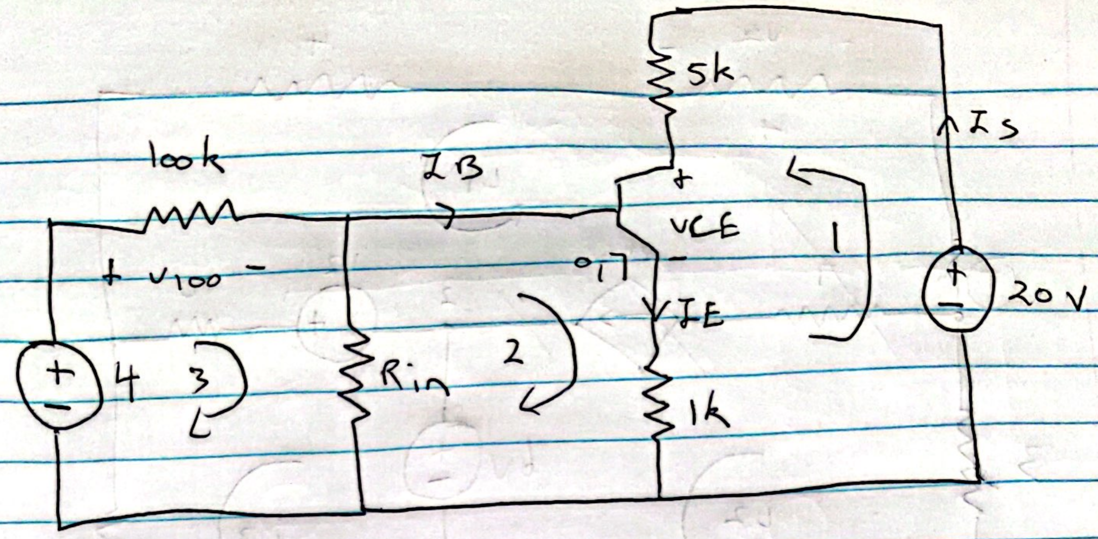
$18I_1 - 10\cancel{I_2} + 6I_2 + 6 + 12 = 0$

I_3

$18I_1 - 10I_3 + 6I_2 = -18$ ②

Loop 3

$14I_3 - 10I_1 = 18$ ③



$I_E = 1{,}01 \text{ mA}$

$\beta = 100$

$V_{BE} = 0{,}7$

$I_E = (1 + 100) I_B$

$1{,}01 \times 10^{-3} = 101 I_B$

$I_B = \cancel{10 \times 10^{-6}}$

10×10^{-6}

$I_S = I_C = \beta I_B$

$= 100 \times (10 \times 10^{-6})$

$= 0{,}001 \text{ A}$

1 KVL

$-20 + 5000 I_C + V_{CE} + 1000 I_E = 0$

2 KVL

$-V_R + 0{,}7 + 1000 I_E = 0$

3 KVL

$-4 + V_{100} + V_R = 0$